

CNG KSP Feature Matrix

Every capability observed across shipping CNG Key Storage Providers, with this project's coverage and — where a gap exists — what closing it would take. Generated 2026-09-09.

97	49	7	41	51%	40
Capabilities tracked	Covered	Partial	Not covered	Fully covered	Actionable gaps

How to read this

Status	Meaning	Ev	Evidence for the market claim
Covered	Implemented and exercised by the test suite.	A	Read from source code or an authoritative API reference that enumerates the provider's own algorithm list.
Partial	Implemented, but reachable only through a non-standard identifier or with a stated caveat.	B	Stated in vendor or Microsoft documentation surfaced through search; primary page not read in full.
Not covered	Absent. The Gap column says what closing it would take.	C	Drawn from HSM datasheets describing the device rather than its KSP. Treat as a lead, not a fact.

Effort on gaps: S = a few lines to a day · M = days · L = a substantial piece of work. **N/A** marks a deliberate position or an external blocker rather than a backlog item — those are excluded from the actionable-gap count.

ID	Feature	Detail	Market support	Ev	Status	Gap — what closing it would take	Eff
PROVIDER INTERFACE							
IFACE-01	GetKeyStorageInterface export	Sole exported symbol; returns the KSP function table	All providers	A	Covered	—	—
IFACE-02	OpenProvider / FreeProvider	Provider handle lifecycle	All providers	A	Covered	—	—
IFACE-03	GetProviderProperty	Name / Version / ImplType	All providers	A	Covered	—	—
IFACE-04	SetProviderProperty	Provider-level configuration	Vendor-specific	C	Not covered	Stubbed as NTE_NOT_SUPPORTED. Would need a defined config surface — slot selection and PIN source. Low value until multi-slot exists.	M
IFACE-05	NotifyChangeKey	Key-change notification handle	Microsoft KSP	B	Not covered	Returns ERROR_SUCCESS as a no-op. Real support needs a registry/file watcher on the token store and an event object per caller.	L
IFACE-06	GetOperationProperty	Async / overlapped operation state	Microsoft KSP	B	Not covered	Returns NTE_NOT_SUPPORTED. Only needed for overlapped I/O; PKCS#11 is synchronous so there is nothing to report.	L
IFACE-07	PromptUser	Interactive PIN / consent dialog	Smart card KSPs	B	Not covered	Returns NTE_NOT_SUPPORTED. Would need a UI thread and a window handle from NCrypt_■WINDOW_■HANDLE_■PROPERTY.	L
IFACE-08	FreeBuffer / FreeObject	Provider-allocated memory release	All providers	A	Covered	—	—
TABLE-01	Function table layout matches the SDK	NCrypt_■KEY_■STORAGE_■FUNCTION_■TABLE slot order and membership	All providers	A	Not covered	The table in ksp_main.c is ordered to a hand-written struct in tests/mock/windows_■compat.h, not to the SDK. It omits IsAlgSupported, EnumAlgorithms and VerifySignature and adds GetOperationProperty and FreeObject, which are not KSP entry points, so every slot from the thirteenth on is displaced. ksp_main.h also includes ncrypt.h, which does not declare this struct at all; the real declaration is in ncrypt_provider.h. Build once against the Windows SDK header and let the compiler assign the slots by name.	M
ASYMMETRIC RSA							

ID	Feature	Detail	Market support	Ev	Status	Gap — what closing it would take	Eff
RSA-01	RSA key generation	2048 / 3072 / 4096 bits	All providers	A	Covered	—	—
RSA-02	RSA 1024 and below	512-1024 bits	MS Software KSP; Utimaco	B	Partial	Build-time opt-in only. KSP_RSA_MIN_BITS defaults to 2048; a legacy build can lower it, but the shipped default still rejects <2048 with NTE_BAD_LEN. Deliberate: silently allowing 1024 would weaken every deployment.	S
RSA-03	RSA above 4096	8192 / 16384 bits	MS Software KSP; Utimaco	B	Covered	—	—
RSA-04	RSA arbitrary size increments	64-bit (MS) or 8-bit (Utimaco) steps	MS Software KSP; Utimaco	B	Covered	—	—
RSA-05	Custom public exponent	Exponent other than 65537	Vendor-specific	C	Covered	—	—
RSA-06	PKCS#1 v1.5 signing	SHA-1 / 256 / 384 / 512	All providers	A	Covered	—	—
RSA-07	PSS signing	SHA-1 / 256 / 384 / 512 with caller salt length	All providers	B	Covered	—	—
RSA-08	PSS with SHA-224	SHA-224 salt and MGF1	Unknown	C	Covered	—	—
RSA-09	PKCS#1 v1.5 decryption	AT_KEYEXCHANGE keys	All providers	A	Covered	—	—
RSA-10	OAEP decryption	SHA-1 / 224 / 256 / 384 / 512	All providers	A	Covered	—	—
RSA-11	OAEP application label	pbLabel / cbLabel passed through	Unknown	C	Covered	—	—
RSA-12	Raw RSA (no padding)	CKM_RSA_X_509	Some HSMs	C	Not covered	SoftHSM2 does not implement CKM_RSA_X_509. Blocked by the backend not by the KSP; would work against an HSM that offers it.	M
ASYMMETRIC ECDSA							
ECDSA-01	ECDSA P-256	secp256r1; 64-byte signature	All providers	A	Covered	—	—
ECDSA-02	ECDSA P-384	secp384r1; 96-byte signature	All providers	A	Covered	—	—
ECDSA-03	ECDSA P-521	secp521r1; 132-byte signature	All providers	A	Covered	—	—
ECDSA-04	Brainpool curves	brainpoolP256r1 / 384r1 / 512r1	Entrust; Utimaco (device)	C	Not covered	Add the DER OIDs to config.h and P11_GetCurveOid. SoftHSM2 supports them. CNG has BCRYPT_■ECC_■CURVE_■BRAINPOOL* names so this stays standards-compliant.	M
ECDSA-05	secp256k1	Bitcoin curve	Entrust (device)	C	Partial	Curve, OID and mechanism are wired and the key generates and signs. Reachable only through this KSP's own ECDSA_SECP256K1 identifier; the standard CNG route is the generic ECDSA algorithm with BCRYPT_■ECC_■CURVE_■NAME set to BCRYPT_■ECC_■CURVE_■SECP256K1, which needs curve-name property handling in ksp_properties.c.	S
ECDSA-06	DER to r s conversion	PKCS#11 returns DER; CNG wants raw	All providers	A	Covered	—	—
KEY AGREEMENT							
ECDH-01	ECDH P-256 / P-384 / P-521	CKM_ECDH1_DERIVE with CKD_NULL	MS Software KSP; Utimaco	B	Covered	—	—
ECDH-02	SecretAgreement / DeriveKey / FreeSecret	NCRYPT_■SECRET_■HANDL E lifecycle	MS Software KSP; Utimaco	B	Covered	—	—
ECDH-03	BCRYPT_KDF_RAW_SECRET	Raw Z value returned	MS Software KSP	B	Covered	—	—
ECDH-04	BCRYPT_KDF_HASH	SP 800-56A concatenation KDF	MS Software KSP	B	Not covered	Returns NTE_NOT_SUPPORTED. Implement by hashing the Z value with the caller-supplied algorithm and info from the NCryptBufferDesc. Moderate work; callers can run it with BCrypt instead.	M
ECDH-05	BCRYPT_KDF_HMAC / TLS_PRF / HKDF	Protocol-specific KDFs	MS Software KSP	B	Not covered	Same shape as ECDH-04. HKDF is the one worth doing first for TLS 1.3.	M
ECDH-06	Finite-field Diffie-Hellman	DH 512-4096	MS Software KSP	B	Not covered	Out of scope by choice. ECDH supersedes it and no surveyed HSM KSP exposes it. Would need CKM_■DH_■PKCS_■KEY_■PAIR_■GEN plus CKM_■DH_■PKCS_■DERIVE.	L

ID	Feature	Detail	Market support	Ev	Status	Gap — what closing it would take	Eff
ASYMMETRIC EDDSA							
EDDSA-01	Ed25519 signing	64-byte raw signature	None	A	Partial	Works against SoftHSM2 but uses the non-standard identifier EDDSA_ED25519. CNG defines no EdDSA algorithm ID so no stock application can request it. Nothing to fix in the KSP; blocked on Microsoft.	N/A
EDDSA-02	Ed448 signing	114-byte raw signature	None	A	Partial	Same constraint as EDDSA-01.	N/A
EDDSA-03	X25519 key agreement	Curve25519 ECDH	MS CNG has BCRYPT_EC_C_CURVE_25519	B	Not covered	This one IS standard in CNG (Windows 10+) unlike Ed25519 signing. Add the curve OID and route it through the existing ECDH path. Highest-value gap in this table.	M
ASYMMETRIC DSA							
DSA-01	DSA signing	512-1024 bits	MS Software KSP	B	Not covered	Deliberately out of scope. Deprecated in modern Windows PKI and absent from every HSM KSP surveyed.	L
POST-QUANTUM							
PQC-01	ML-DSA (Dilithium)	FIPS 204	Thales Luna (device claim)	C	Not covered	SoftHSM2 2.7.0 has no PQC mechanisms so this is blocked at the backend. CNG gained PQC identifiers in recent Windows builds; revisit when both sides support it.	L
PQC-02	ML-KEM (Kyber)	FIPS 203	Thales Luna (device claim)	C	Not covered	Same blocker as PQC-01.	L
PQC-03	LMS / XMSS	Stateful hash-based signatures	Thales Luna (device claim)	C	Not covered	Same blocker as PQC-01. Stateful schemes also need key-state persistence the KSP has no model for.	L
SYMMETRIC AES							
AES-01	AES key generation	128 / 192 / 256 bits	None via KSP	A	Covered	—	—
AES-02	AES-ECB	No IV	None via KSP	A	Covered	—	—
AES-03	AES-CBC	16-byte IV	None via KSP	A	Covered	—	—
AES-04	AES-CBC with padding	NCRYPT_■PAD_■CIPHER_■F LAG	None via KSP	A	Covered	—	—
AES-05	AES-GCM	12-byte nonce; 128-bit tag; AAD	None via KSP	A	Covered	—	—
AES-06	AES-CTR	16-byte counter block	None via KSP	A	Partial	Works but ChainingModeCTR is a KSP extension. CNG defines no standard counter-mode string so no stock application will select it.	N/A
AES-07	AES-CCM	Authenticated mode	None via KSP	A	Not covered	Returns NTE_NOT_SUPPORTED. SoftHSM2 2.7.0 does not implement CKM_AES_CCM so this is a backend blocker not a KSP one.	M
AES-08	AES-CFB	Cipher feedback	None via KSP	A	Not covered	Returns NTE_NOT_SUPPORTED. No SoftHSM2 mechanism; would need CKM_AES_CFB128 which 2.7.0 lacks.	M
AES-09	AES key wrap	RFC 3394 / 5649	Some HSMs (device)	C	Not covered	SoftHSM2 offers CKM_AES_KEY_WRAP and KEY_WRAP_PAD. Needs KSP_ExportKey / ImportKey wired to C_WrapKey / C_UnwrapKey plus a CNG blob convention.	L
AES-10	AES-CMAC	Block-cipher MAC	Some HSMs (device)	C	Not covered	SoftHSM2 offers CKM_AES_CMAC. Would route through KSP_SignHash like HMAC does; same non-standard-identifier caveat applies.	M
SYMMETRIC MAC							
HMAC-01	HMAC-SHA1 / 256 / 384 / 512	Generic secret keys	None via KSP	A	Partial	Works against SoftHSM2 but HMAC_SHA* are our own identifiers. CNG does HMAC through BCrypt algorithm handles not KSP key algorithms.	N/A
HMAC-02	HMAC-SHA224	SHA-224 variant	None	C	Covered	—	—
SYMMETRIC LEGACY							
3DES-01	3DES / DES	56 / 112 / 168 bits	None via KSP	B	Not covered	Deliberately out of scope. Deprecated and excluded from modern HLK requirements despite SoftHSM2 supporting 14 DES mechanisms.	L
KEY FORMATS							

ID	Feature	Detail	Market support	Ev	Status	Gap — what closing it would take	Eff
FMT-01	BCRYPT_RSAPUBLIC_BLOB export	Header + exponent + modulus	All providers	A	Covered	—	—
FMT-02	BCRYPT_ECCPUBLIC_BLOB export	Header + X + Y	All providers	A	Covered	—	—
FMT-03	EdDSA public blob export	Generic ECC magic + raw point	None	A	Covered	—	—
FMT-04	BCRYPT_RSAPUBLIC_BLOB import	Creates a session public key object	Vendor-specific	C	Covered	—	—
FMT-05	BCRYPT_ECCPUBLIC_BLOB import	Creates a session public key object	Vendor-specific	C	Covered	—	—
FMT-06	Private key export	RSAPUBLICPRIVATE / ECCPRIVATE	None (all refuse)	A	Not covered	Refused by design with NTE_NOT_SUPPORTED. Keys are CKA_■EXTRACTABLE=FALSE. This is correct HSM behaviour not a gap.	N/A
FMT-07	Private key import	Wrapping an external private key onto the token	Some HSMs	C	Not covered	Returns NTE_NOT_SUPPORTED. Would need C_CreateObject with private material or C_UnwrapKey. Contradicts the HSM posture; only add if a migration workflow demands it.	L
FMT-08	BCRYPT_KEY_DATA_BLOB	Raw symmetric key material	MS Software KSP	B	Not covered	AES keys are created CKA_■EXTRACTABLE=FALSE so export cannot work. Import would need C_CreateObject with CKA_VALUE. Constants are already in the mock header.	M
FMT-09	PKCS#8 blob	NCRYPT_■PKCS8_■PRIVATE_■KEY_■BLOB	MS Software KSP	B	Not covered	Same constraint as FMT-07: it is a private-key format and keys are non-extractable.	L
FMT-10	OPAQUETRANSFER blob	Vendor-opaque key transport	Vendor-specific	C	Not covered	Would let keys move between instances of the same provider. Needs a defined wrapping format; no standard to follow.	L
KEY PROPERTIES							
PROP-01	NCRYPT_ALGORITHM_PROPERTY	Read	All providers	A	Covered	—	—
PROP-02	NCRYPT_LENGTH_PROPERTY	Read and write before finalize	All providers	A	Covered	—	—
PROP-03	NCRYPT_KEY_TYPE_PROPERTY	AT_SIGNATURE / AT_KEYEXCHANGE	All providers	A	Covered	—	—
PROP-04	NCRYPT_NAME / UNIQUE_NAME	Key label	All providers	A	Covered	—	—
PROP-05	NCRYPT_EXPORT_POLICY_PROPERTY	Always 0 (non-exportable)	All providers	A	Covered	—	—
PROP-06	NCRYPT_KEY_USAGE_PROPERTY	Signing / decrypt / key agreement flags	All providers	A	Covered	—	—
PROP-07	NCRYPT_ALGORITHM_GROUP_PROPERTY	RSA / ECDSA / ECDH / EDDSA / AES / HMAC	All providers	A	Covered	—	—
PROP-08	NCRYPT_CHAINING_MODE_PROPERTY	Read and write on symmetric keys	MS Software KSP (BCrypt)	A	Covered	—	—
PROP-09	NCRYPT_INITIALIZATION_VECTOR	Read and write on symmetric keys	MS Software KSP (BCrypt)	A	Covered	—	—
PROP-10	NCRYPT_BLOCK_LENGTH_PROPERTY	AES block size	MS Software KSP	B	Covered	—	—
PROP-11	NCRYPT_SECURITY_DESCRIPTOR_PROPERTY	Per-key ACL	MS Software KSP	B	Not covered	Returns NTE_NOT_SUPPORTED. PKCS#11 has no ACL model; would need a side-channel store mapping key labels to SDDL strings.	L
PROP-12	NCRYPT_UI_POLICY_PROPERTY	Strong-key protection prompt	MS Software KSP; smart cards	B	Not covered	Depends on PromptUser (IFACE-07). Same UI-thread requirement.	L

ID	Feature	Detail	Market support	Ev	Status	Gap — what closing it would take	Eff
PROP-13	NCRYPT_PIN_PROPERTY	Per-key PIN	Smart card KSPs	B	Not covered	PIN is process-wide from SOFTHSM2_PIN. Per-key PINs would need a per-handle credential cache and re-login per operation.	M
PROP-14	NCRYPT_CERTIFICATE_PROPERTY	Certificate stored alongside the key	Smart card KSPs	B	Not covered	Would store a CKO_CERTIFICATE object with the same CKA_LABEL. SoftHSM2 supports certificate objects so this is straightforward.	M
PROP-15	NCRYPT_WINDOW_HANDLE_PROPERTY	Parent HWND for dialogs	Smart card KSPs	B	Not covered	Only meaningful with PromptUser (IFACE-07).	L
PROP-16	NCRYPT_USE_COUNT_PROPERTY	Key usage counter	Some HSMs	C	Not covered	Would need a counter object on the token incremented per operation. SoftHSM2 has no atomic counter primitive.	L
KEY LIFECYCLE							
LIFE-01	CreatePersistedKey	Immediate and deferred generation	All providers	A	Covered	—	—
LIFE-02	FinalizeKey	Deferred generation trigger	All providers	A	Covered	—	—
LIFE-03	OpenKey by name	Asymmetric and symmetric	All providers	A	Covered	—	—
LIFE-04	DeleteKey	Destroys all token objects	All providers	A	Covered	—	—
LIFE-05	EnumKeys	Paged enumeration with state	All providers	A	Covered	—	—
LIFE-06	EnumAlgorithms	NCryptEnumAlgorithms support	All providers	B	Not covered	Re-graded. Earlier note was wrong: this is not served from the registry. NCRYPT_KEY_STORAGE_FUNCTION_TABLE has a real EnumAlgorithms slot that this KSP does not fill, and registration must also publish the list via BCryptAddContextFunctionProvider. Needs the function-table correction in TABLE-01 first.	M
LIFE-07	IsAlgSupported	NCryptIsAlgSupported	All providers	B	Not covered	Re-graded, same cause as LIFE-06: IsAlgSupported is a function-table slot, not a registry lookup. Blocked on TABLE-01.	M
LIFE-08	Machine vs user key scope	NCRYPT_MACHINE_KEY_FLAG	All providers	B	Not covered	All keys are token-global. PKCS#11 has no per-user scoping within a slot; would need label prefixing plus an ACL model.	M
OPERATIONAL							
OPS-01	Hardware key protection	Tamper-resistant storage	All commercial providers	A	Not covered	Fundamental: SoftHSM2 is a software token using encrypted SQLite. Only fixable by pointing the KSP at a real HSM which the PKCS#11 abstraction already allows.	N/A
OPS-02	FIPS 140-2/3 validation	Certified cryptographic module	All commercial providers	B	Not covered	Follows from OPS-01. Requires certified hardware and a formal validation programme.	N/A
OPS-03	Authenticode-signed binary	Commercial code-signing certificate	All commercial providers	A	Not covered	Buy an OV or EV certificate and sign with signtool. No Microsoft involvement needed. See doc 10.	S
OPS-04	Multiple slots / tokens	Selecting among several tokens	All commercial providers	B	Not covered	Only the first token-present slot is used. Add a SOFTHSM2_SLOT environment variable or a provider property and thread the slot ID through P11_CONTEXT.	M
OPS-05	Key attestation	TPM / HSM-signed key origin proof	MS Platform Crypto; some HSMs	B	Not covered	No PKCS#11 standard mechanism exists. Vendor-specific and unreachable through a portable PKCS#11 layer.	L
OPS-06	Load balancing / HA	Multiple HSM members	Thales; Entrust	C	Not covered	Handled inside vendor PKCS#11 libraries. Our KSP would inherit it automatically when pointed at such a library; nothing to implement.	N/A
OPS-07	Session pool concurrency	Concurrent NCrypt calls	All providers	A	Covered	—	—
OPS-08	Re-initialisation without restart	Recover from a bad DLL path	Vendor-specific	C	Not covered	InitOnceExecuteOnce is one-shot per process even on failure. Would need a resettable init guard; a real fix requires care around in-flight sessions.	M
OPS-09	PIN caching / re-login	Session recovery after token logout	All commercial providers	C	Partial	C_Login tolerates CKR_USER_ALREADY_LOGGED_IN but there is no re-login path if a session drops. Add a retry that reopens and re-logs in on CKR_USER_NOT_LOGGED_IN.	M
OPS-10	Logging / diagnostics	Trace output for support	All providers	A	Covered	—	—

Maintaining this document

The source of truth is `softhsm_ksp/docs/feature-matrix.csv`. One row per capability. To record that a gap has been closed, change that row's *status* to Covered and clear its *gap_solution*; to add a newly observed market feature, append a row with a fresh ID. Then regenerate:

```
python3 generate_feature_matrix.py
```

Every number on page 1 is computed from the CSV at generation time, so the summary cannot fall out of step with the table. Because the CSV is plain text, a change to coverage shows up as a one-line diff in review rather than an opaque binary change.

Actionable gaps by effort: **3 small, 19 medium, 18 large**. Rows marked N/A are excluded — they are deliberate positions (private key export is refused by design) or external blockers (CNG defines no EdDSA identifier; SoftHSM2 is a software token).

A caution on the market columns. An HSM datasheet is not a statement about its CNG provider. AWS CloudHSM performs AES, DES3 and HMAC in hardware, yet its KSP answers `NCryptIsAlgSupported` for exactly four identifiers. Every grade-C cell here carries that uncertainty. See `docs/11-market-comparison.md` for the full audit and its sources.